Part 1: From a Box of Hot Gas to an Escaping Rocket

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We design a rocket engine from first principles and use it to launch a spacecraft off its home planet. We model the engine as many tiny boxes of hot hydrogen gas ($T = 3000\,\mathrm{K}$), each with a small nozzle in one wall. Our own molecular simulation of a box reproduces the Maxwell–Boltzmann predictions for speed, energy and pressure to within the statistical error expected for $N = 1 \times 10^5$ particles. With a nozzle and a fuel inlet, one box gives a thrust of $5.759 \times 10^{-10}\,\mathrm{N}$ for a fuel use of $1.317 \times 10^{-13}\,\mathrm{kg/s}$, so the exhaust speed is $u = 4374\,\mathrm{m/s}$. We then design the launch: 7.976×10^{15} boxes give 4,594 kN of thrust and burn 1050.1 kg/s. A thrust-to-weight ratio of 1.041 at liftoff gives a 6.00-minute radial launch that reaches escape velocity 686 km up, with 10,000 kg of fuel left for the journey. Gravity takes 25% of the rocket equation's ideal Δv. The final position in the star's frame was verified by `ast2000tools` (verification passed, solar system seed 42716).

I. INTRODUCTION

A rocket works by throwing mass backwards. The faster the exhaust leaves, the less fuel a given change in speed costs, and the exhaust speed of a chemical or thermal engine is set by how fast its gas molecules move. In this first part of the AST2000 spacecraft project [1] we follow this chain from the bottom up.

- Statistics of hot gas: the Gaussian and Maxwell–Boltzmann distributions (Section II).
- A molecular simulation of a closed box of hydrogen, tested against those predictions (Section III).
- A nozzle in that box: thrust and fuel use per box (Section IV).
- The rocket equation, and a full launch simulation with gravity, used to size the engine (Section V).
- Converting the result to the star's frame, checked by the official `ast2000tools` package [2] (Section VI).

We aimed to understand every number well enough to predict how the result changes if something goes wrong. So beyond the required results we report error budgets, sensitivities and a full energy accounting of the launch.

II. GAS STATISTICS

A. Theory

A normal distribution with mean μ and standard deviation σ has the density

$$f(\mu, \sigma; x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]. \tag{1}$$

In an ideal gas at temperature T, each velocity component of a molecule of mass m follows this distribution with $\mu = 0$ and $\sigma_v = \sqrt{kT/m}$ [3]. The three components are independent, and the speed $v = |\mathbf{v}|$ then follows the Maxwell–Boltzmann distribution

$$P(v) = \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT}\, 4\pi v^2. \tag{2}$$

Integrating Eq. (2) gives the mean speed $\langle v\rangle = \sqrt{8kT/\pi m}$, the mean kinetic energy $\langle E\rangle = \frac{3}{2}kT$ and the pressure $P = nkT$.

B. Method

We wrote a composite Simpson integrator, whose error falls as n^{-4} with the number of intervals n. We tested it on the probabilities of falling within 1, 2 and 3σ and on the full width at half maximum (FWHM). We then evaluated $\langle v\rangle$, $\langle E\rangle$ and P numerically for H_2 at $T = 3000\,\mathrm{K}$.

C. Results

The integrator gives $P(|x-\mu| < k\sigma) = 0.6827$, 0.9545 and 0.9973 for $k = 1, 2, 3$, and FWHM $= 2.3548\,\sigma$, which equals $2\sqrt{2\ln 2}\,\sigma$ (Fig. 1). For H_2 at 3000 K, $\sigma_v = 3518\,\mathrm{m/s}$. The most probable, mean and rms speeds are 4975, 5613 and 6093 m/s (Fig. 2). Only a fraction 0.0776 of the molecules have $5 \le v_x \le 30\,\mathrm{km/s}$, which is about 7760 of 10^5.

One point looks like a paradox: the velocity components peak at zero, but the speed distribution peaks at $v_{mp} > 0$. Figure 3 resolves it. Velocity *vectors* are densest at the origin, but the shell of speeds near v grows as $4\pi v^2$.

A finite gas only estimates these averages. The relative scatter of a sample mean is $\sqrt{3\pi/8 - 1}/\sqrt{N}$ for $\langle v\rangle$ and $\sqrt{2/3}/\sqrt{N}$ for $\langle E\rangle$ and P (Fig. 4). For $N = 10^5$ these are 0.13% and 0.26%. This sets the accuracy we can expect from the box simulation.

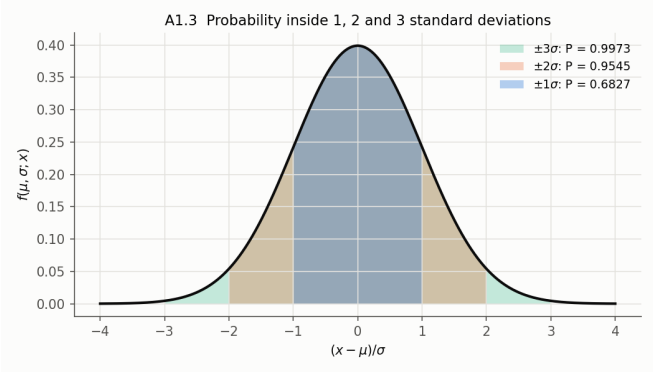


FIG. 1. Probability within 1, 2 and 3σ of the mean, from our Simpson integrator.

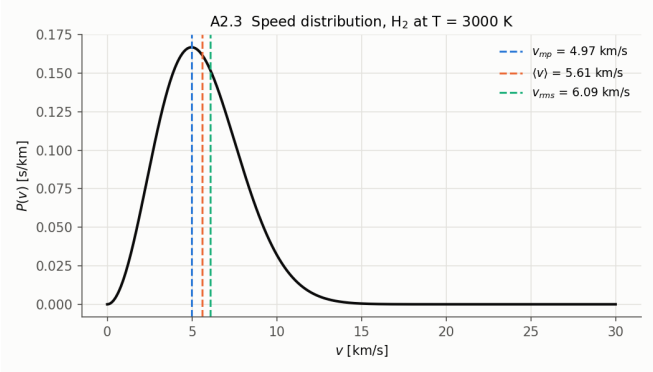


FIG. 2. Maxwell–Boltzmann speed distribution of H_2 at 3000 K, with the three characteristic speeds.

III. A BOX OF GAS

A. Method

We place $N = 1 \times 10^5$ H_2 molecules uniformly in a cube of side $L = 1\,\mu$m, with Maxwell–Boltzmann velocities. They move freely and bounce elastically off the walls, and we follow them for $\tau = 1$ ns.

Between bounces the motion is an exact straight line, and a bounce is an exact mirror image. A molecule that crosses a wall during a step is therefore reflected back through it ($x \to -x$ or $x \to 2L - x$) and its normal velocity is reversed. This puts it exactly where a mid-step bounce would have. The simulation is then *exact for any time step*, as long as no molecule crosses the whole box in one step. The only errors left are statistical. Each step only about 1% of the molecules reach a wall, so the bookkeeping runs on that subset. The pressure is the momentum given to the walls, $2m|v_\perp|$ per hit, divided by area and time.

B. Results

The velocity and position distributions stay Maxwellian and uniform throughout the run (Fig. 5). Figure 6 compares every measured quantity with theory. The deviations are 0.147% ($\langle v \rangle$), 0.276% ($\langle E \rangle$), 0.317% (P) and 0.105% (wall hit rate). Each is of the size the $1/\sqrt{N}$ noise predicts (0.26% for P).

Figure 7 shows why the deviations look the way they do. All six walls see the same molecules for the whole run, and $|v_i|$ never changes in a bounce. Each wall therefore measures *this sample's* $\langle v_i^2 \rangle$, not the ensemble value kT/m. Predicting each wall from the sample's own velocities reproduces the measured pressures to 0.17%, which is the counting noise of a 1 ns run. The convergence study (Fig. 8) confirms both points. The pressure error falls as $1/\sqrt{N}$, and it does not depend on the time step at all.

IV. THE ENGINE

A. Theory

A hole of area A in a wall of a gas in equilibrium lets out $nA\sqrt{kT/2\pi m}$ particles per second. They carry momentum $PA/2$ per second, so the effusion exhaust speed is $u = \sqrt{\pi kT/2m} = 4409$ m/s.

B. Method

We cut a square hole of area $0.25L^2$ in the middle of the wall $x = L$ and count the momentum mv_x of every molecule that leaves. To keep N constant, each escaped molecule must be replaced, and *how* it is replaced turns out to matter. We use a fuel inlet the size of the hole in the opposite wall. It injects molecules with the flux-weighted (Rayleigh) v_x distribution that a wall at temperature T emits. We compare this with three alternatives:

- a closed box (equilibrium);
- reinjecting the same molecule at the opposite wall;
- a fresh Maxwellian molecule at a random spot.

Thrust and fuel use are averaged over 2 ns after a 0.5 ns burn-in.

C. Results

The engine box settles into a steady flow toward the nozzle within the burn-in, and its temperature stays at 3000 K (Fig. 9). Per box we find

- thrust $F = 5.759 \times 10^{-10}$ N;
- fuel use $\dot{m} = 1.317 \times 10^{-13}$ kg/s;

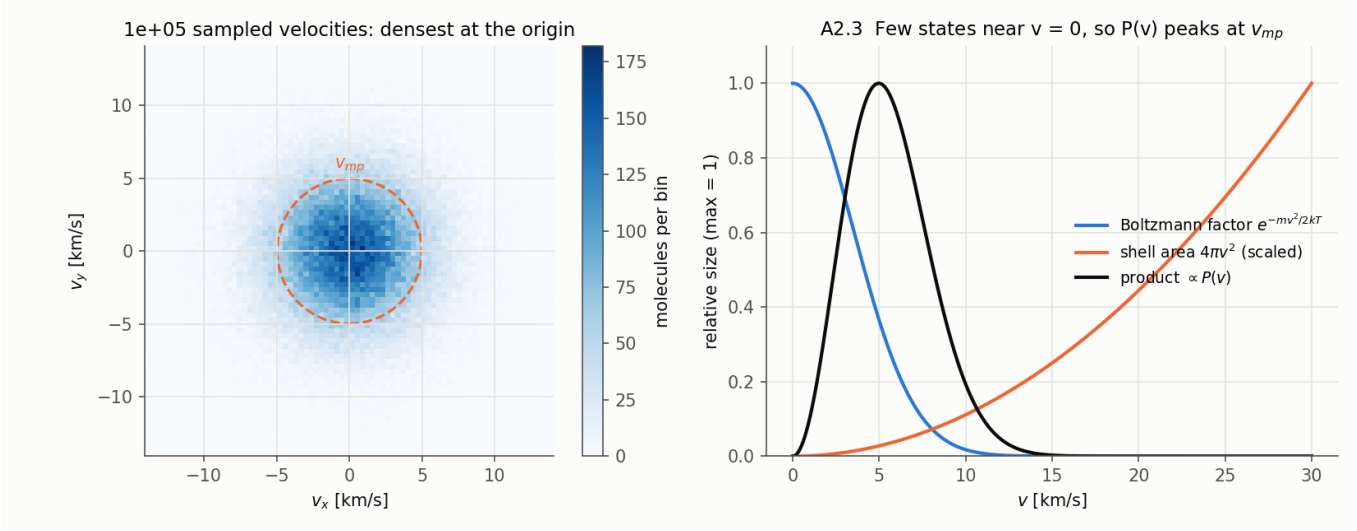


FIG. 3. Left: 10^5 sampled velocity vectors are densest at the origin. Right: the Boltzmann factor falls with v while the area of a speed shell grows as v^2. Their product is $P(v)$, which peaks at v_{mp}.

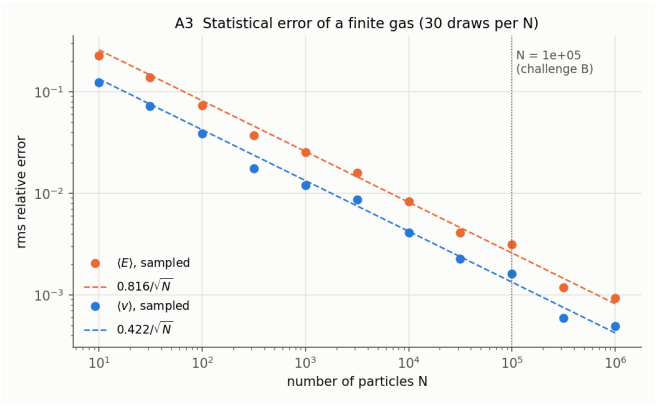


FIG. 4. RMS relative error of $\langle v \rangle$ and $\langle E \rangle$ for a gas of N particles, with the expected $1/\sqrt{N}$ laws.

- exhaust speed $u = F/\dot{m} = 4374\,\mathrm{m/s}$.

These come from 58,997 escaped molecules, so the Poisson error is 0.41%. The thrust is 1.112 times the equilibrium value. The inlet feeds a beam aimed at the nozzle, so more molecules leave than in equilibrium (Fig. 11). The exhaust speed, however, is almost unchanged, since it is set by the flux-weighted velocity distribution (Fig. 10).

The refill method matters (Fig. 12), with thrust ratios

- 0.991 for the closed box;
- 1.112 for the inlet;
- 1.151 for reinjection at the opposite wall;
- 0.775 for random refill.

Random refill fails because the fastest molecules escape most often but are replaced by average ones. This cools the gas to 2617 K and the thrust keeps falling.

V. LAUNCH

A. The rocket equation

Without gravity, momentum conservation gives $m\,\mathrm{d}v = -u\,\mathrm{d}m$, so a boost Δv costs

$$\Delta m = m_0 \left(1 - e^{-\Delta v/u}\right) \tag{3}$$

of fuel. Equation (3) depends only on u, not on the number of boxes. The engine size only sets how *fast* the boost happens (Fig. 13). A step-by-step simulation of a boost agrees with Eq. (3) to 10^{-5}.

B. Method

The rocket rises radially from the equator of the home planet, which has

- radius $R = 6,592\,\mathrm{km}$;
- mass $M = 7.379 \times 10^{24}\,\mathrm{kg}$;
- surface gravity $g_0 = 11.34\,\mathrm{m/s^2}$;
- escape speed $12.22\,\mathrm{km/s}$;
- rotation period 30.3 h, which gives an equatorial speed of $v_{rot} = 379\,\mathrm{m/s}$.

We integrate (r, v_r, m) with

$$\dot{v}_r = \frac{F}{m} - \frac{GM}{r^2}, \qquad \dot{m} = -\dot{m}_{\mathrm{engine}}, \tag{4}$$

using an adaptive Runge–Kutta solver. The solver detects the escape condition $\frac{1}{2}(v_r^2 + v_{rot}^2) = GM/r$ as an exact event.

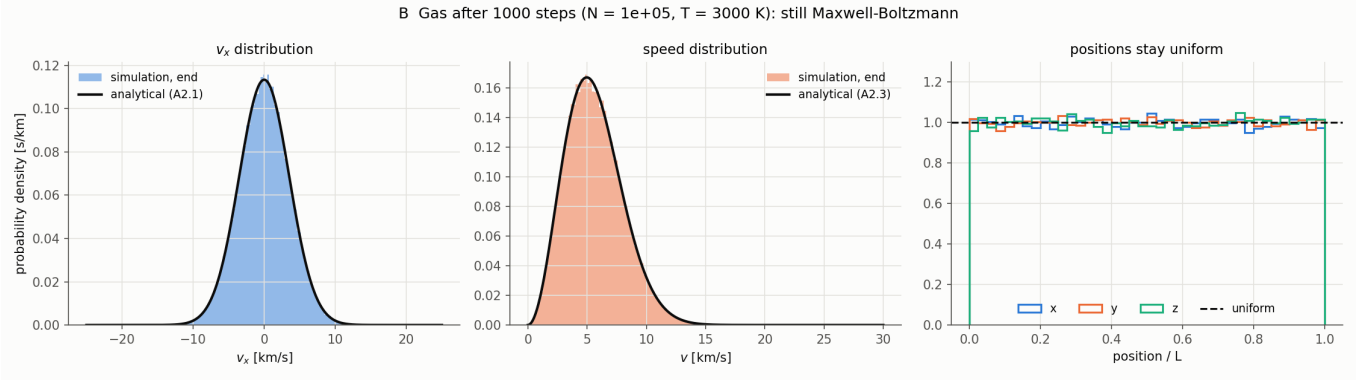


FIG. 5. The gas after the run: v_x and speed distributions against the analytical Maxwell–Boltzmann curves, and uniform positions.

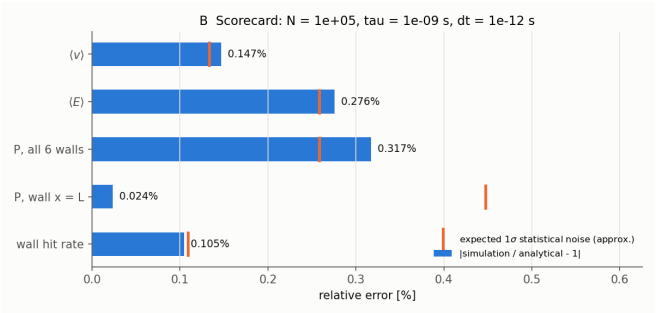


FIG. 6. Scorecard: relative error of each check (bars) against the expected 1σ statistical noise (ticks).

Dividing the equations by F shows that, for a fixed u, the whole launch depends on one number: the initial thrust acceleration F/m_0. So we solve a single 1D root-finding problem for the F/m_0 that gives a 6-minute launch. We then scale the rocket so that exactly 10,000 kg of fuel are left at escape.

C. Results

The design (Fig. 14) uses

- 7.976×10^{15} boxes, giving $F = 4,594$ kN and $\dot{m} = 1050.1$ kg/s;
- 388.0 t of fuel, for a liftoff thrust-to-weight ratio of 1.041.

Escape comes after 360.0 s at a height of 686 km, at $v_r = 11.63$ km/s. By then the launch has burned 378.0 t of fuel. Figure 15 shows why the launch cannot be much longer. A gentler liftoff (thrust-to-weight ratio near 1) lengthens the launch, but it also wastes more fuel on hovering. With $u = 4374$ m/s, about 6.25 minutes is the physical maximum. A hotter engine would allow longer launches. Figure 16 accounts for the fuel. The rocket equation promises $u \ln(m_0/m) = 15.56$ km/s, of which the rocket keeps 11.63 km/s. The remaining $\int g\,\mathrm{d}t = 3.93$ km/s (25%) is the gravity loss. Without gravity the same final speed would have cost 361.9 t of fuel. The crew feels $1.041\,g$ at liftoff, rising steeply as the rocket empties.

Figure 17 asks what happens if the engine is a few percent off. The engine's 0.41% statistical uncertainty changes the reserve by only $\sim 10^2$ kg, a margin we can afford. A 5% weaker engine takes 6.32 min and leaves 9,458 kg. An error in the exhaust speed costs fuel about three times faster than the same error in thrust.

VI. THE STAR'S FRAME

During a launch of a few minutes the planet moves along an essentially straight line. The final state relative to the star is therefore

$$\mathbf{r} = \mathbf{r}_p + (\mathbf{v}_p + v_{rot}\hat{\mathbf{e}}_t)\,\Delta t + r_{\text{final}}\,\hat{\mathbf{e}}_r, \tag{5}$$

$$\mathbf{v} = \mathbf{v}_p + v_{rot}\hat{\mathbf{e}}_t + v_r\hat{\mathbf{e}}_r, \tag{6}$$

where $\hat{\mathbf{e}}_r$ points from the planet's centre to the launch site and $\hat{\mathbf{e}}_t$ points along the rotation. Launching from $\phi = 0$ at $t = 0$ gives the following state at $t = 1.1408 \times 10^{-5}$ yr:

- position (0.68814395, 0.00009498) AU;
- velocity (2.45293, 8.32573) AU/ yr.

`ast2000tools` launched the same rocket with its own 0.01 s Euler steps. It reached escape after 360.01 s and accepted our position (Fig. 18).

The launch site is a free parameter, and Fig. 19 shows that it matters. The planet moves along its orbit at 39.1 km/s. Launching along that motion gives up to 50.72 km/s relative to the star, against 27.45 km/s backwards. Part 5 uses this: it picks the launch site so that "up" points along the departure direction it needs.

VII. DISCUSSION

The main approximations are as follows:

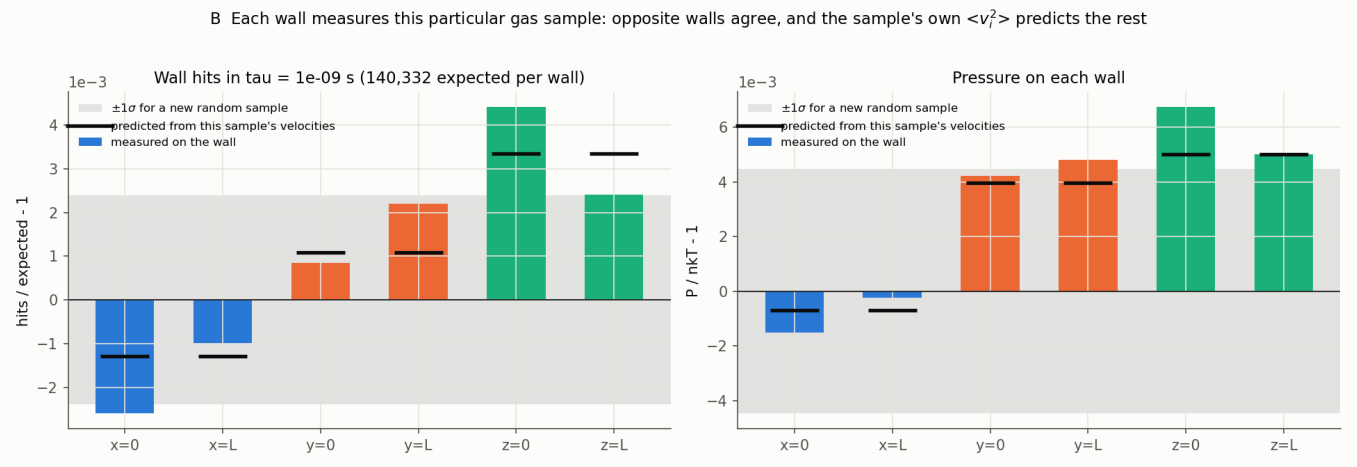


FIG. 7. Every wall separately. Bars: measured hits and pressure. Black ticks: predicted from the velocities of this particular sample. Grey band: the spread expected for a new random sample.

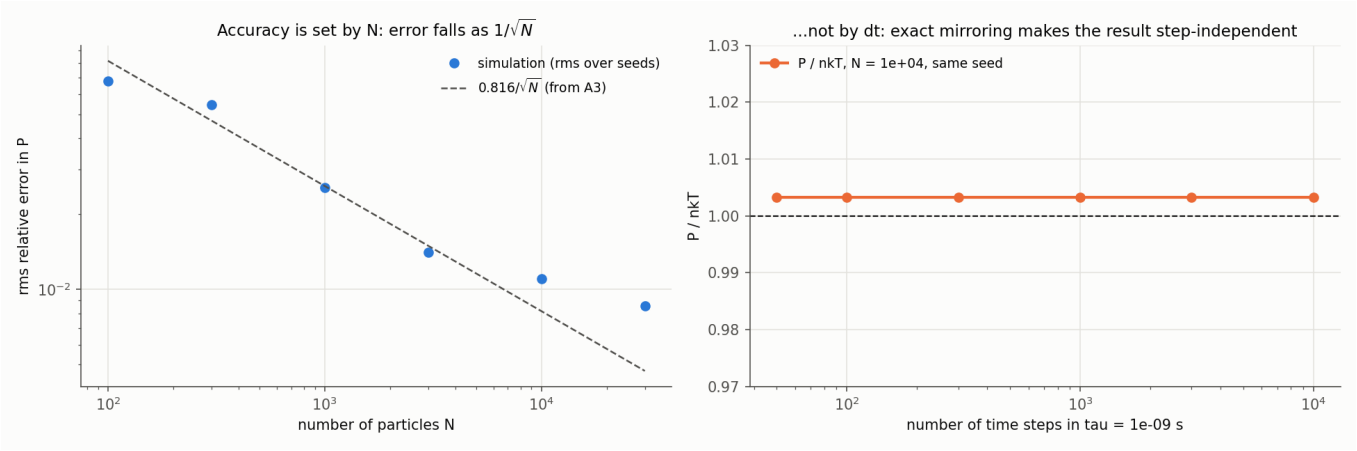


FIG. 8. Left: the pressure error falls as $1/\sqrt{N}$. Right: thanks to exact mirroring, the result does not depend on the time step.

- ideal-gas molecules without collisions;
- a box much smaller than any real engine (we simply take many of them);
- a purely radial launch with no air resistance.

The gas model enters the launch only through two numbers, F and $\dot{m}$ per box, and their ratio u. Figure 17 shows how strongly the reserve depends on them.

The most useful insight is the scaling argument: the launch depends only on F/m_0 and u. It turned the engine design into a one-dimensional root search. It also shows directly that a hotter engine (larger u) is the way to longer, cheaper launches.

The 25% gravity loss is inherent in a radial launch: the rocket spends minutes simply holding itself up. A real rocket turns sideways early to reduce it.

VIII. CONCLUSION

From a box of 10^5 simulated hydrogen molecules we built an engine with an exhaust speed of 4374 m/s. We sized it to lift a 1100 kg spacecraft to escape velocity in 6.00 minutes with 10,000 kg of fuel to spare. The result was verified by `ast2000tools`, and the mission state is saved for the next parts.

ACKNOWLEDGMENTS

Simulations use NumPy and SciPy, and the verification uses `ast2000tools` [2]. The code is in `Project 1/` of the repository, and `pro_1.py` reproduces every figure and number in this paper.

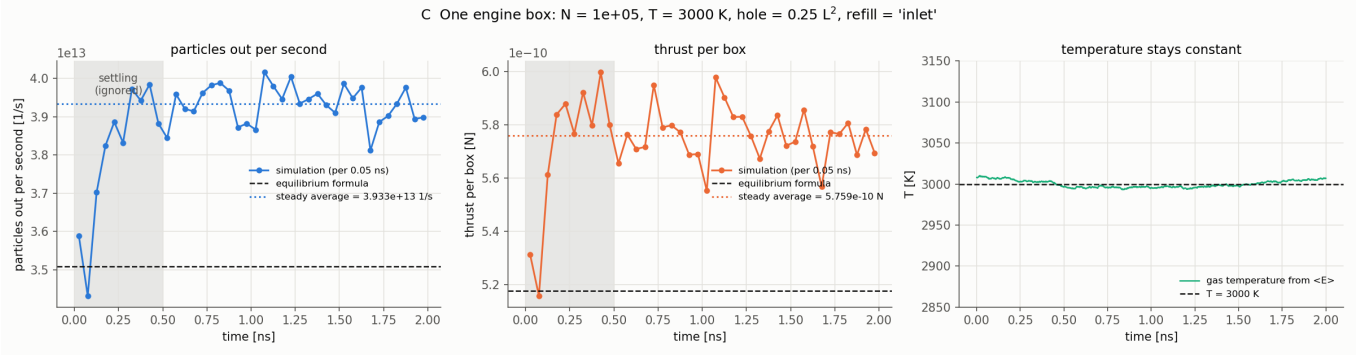


FIG. 9. One engine box. Escape rate, thrust and gas temperature against time. The grey band is the burn-in, which we ignore.

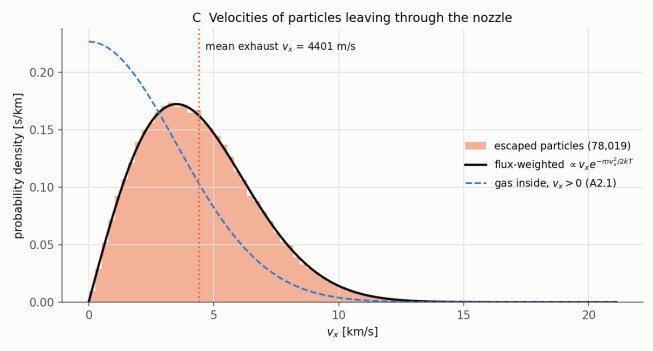


FIG. 10. The escaping molecules follow the flux-weighted distribution $\propto v_x e^{-mv_x^2/2kT}$, not the distribution inside the gas.

[1] AST2000 teaching team, AST2000 spacecraft project, parts 1–9, Institute of Theoretical Astrophysics, University of Oslo (2023).
[2] F. K. Hansen and L. F. Engedal, ast2000tools: Python package for the AST2000 project, https://frodekh.github.io/ast2000tools/html/index.html (2023).
[3] D. V. Schroeder, *An Introduction to Thermal Physics* (Addison Wesley Longman, San Francisco, 2000).

Appendix A: Mission log

Part 1 key numbers (seed 42716)	
Engine temperature	3000 K
Exhaust speed u	4374 m/s
Engine boxes	7.976×10^{15}
Thrust / fuel use	4,594 kN / 1050.1 kg/s
Fuel loaded / left	388.0 t / 10,000 kg
Launch duration	360.0 s
Height / speed at escape	686 km / 11.63 km/s
Gravity loss	3.93 km/s
API verification	passed

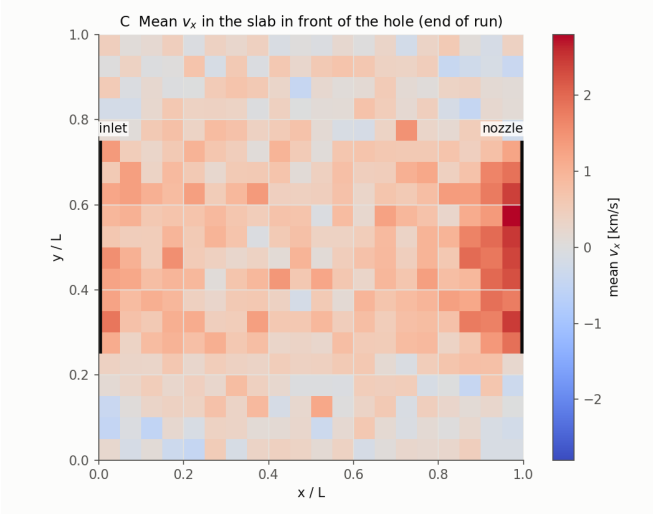


FIG. 11. Mean v_x in the slab in front of the hole: the gas streams from the inlet toward the nozzle.

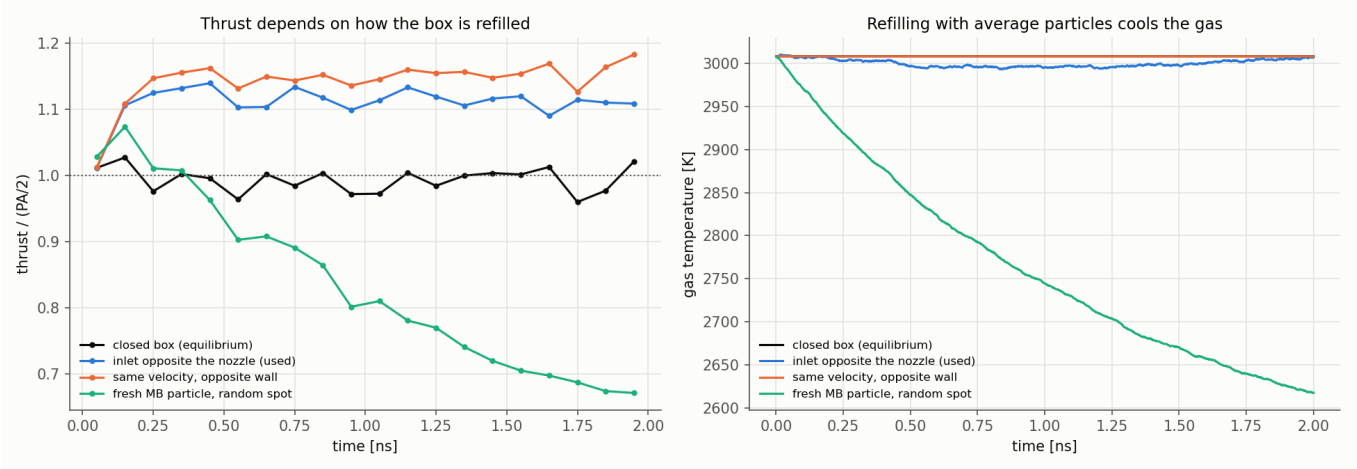


FIG. 12. Thrust and temperature for four ways of keeping N constant. Only the inlet keeps both steady.

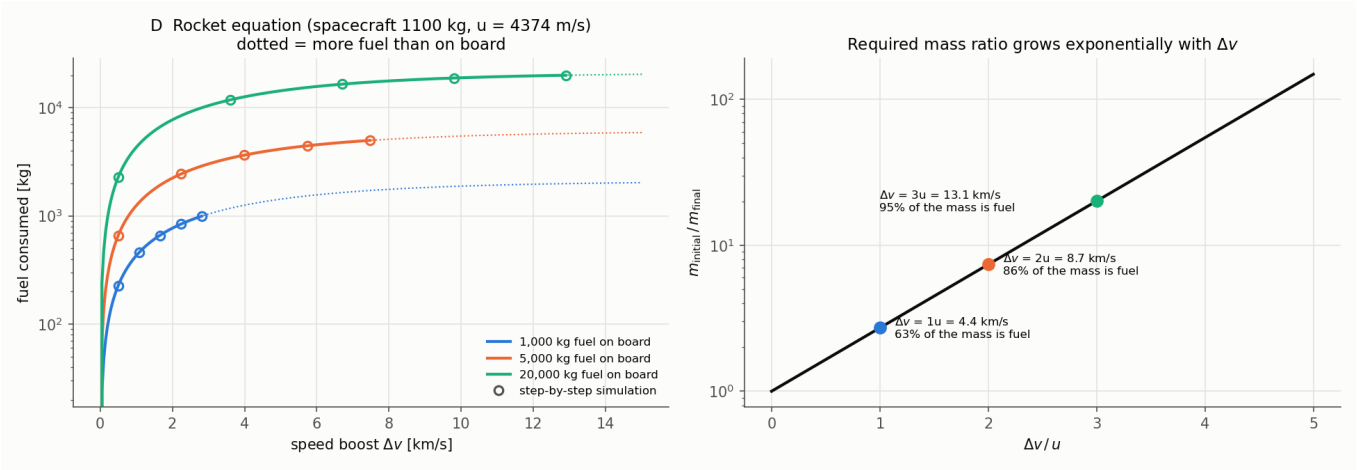


FIG. 13. Left: fuel for a boost against Δv, from Eq. (3) (lines) and a step-by-step simulation (circles). Right: the required mass ratio grows exponentially with $\Delta v/u$.

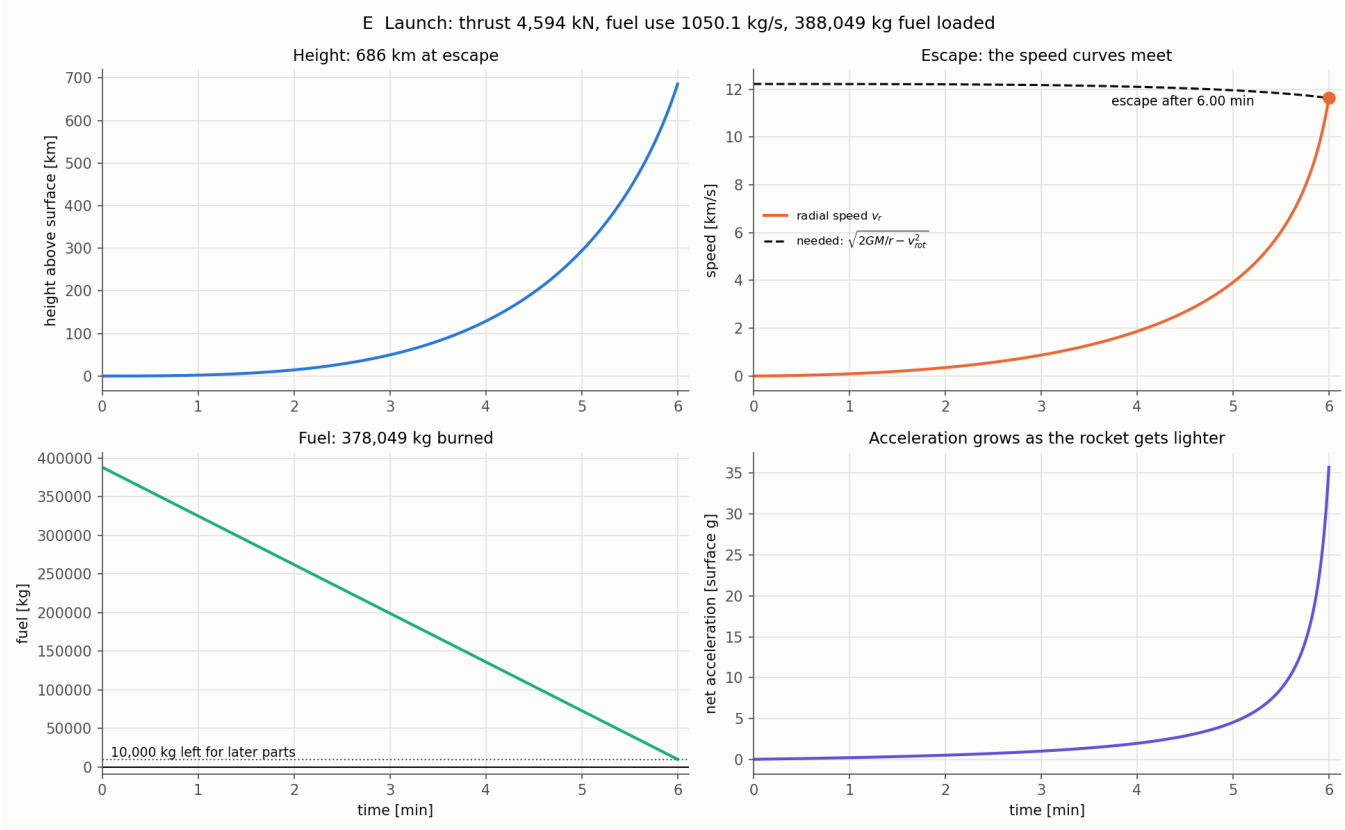


FIG. 14. The launch: height, radial speed against the escape speed, fuel and net acceleration.

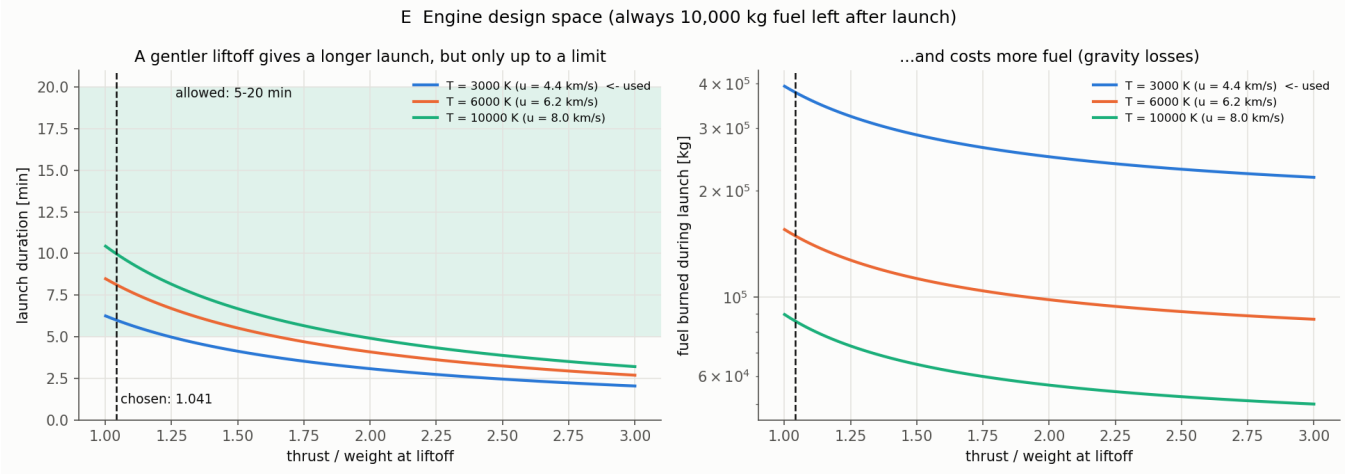


FIG. 15. Engine design space. Launch duration and fuel burned against the thrust-to-weight ratio at liftoff, for three engine temperatures. The dashed line is our design.

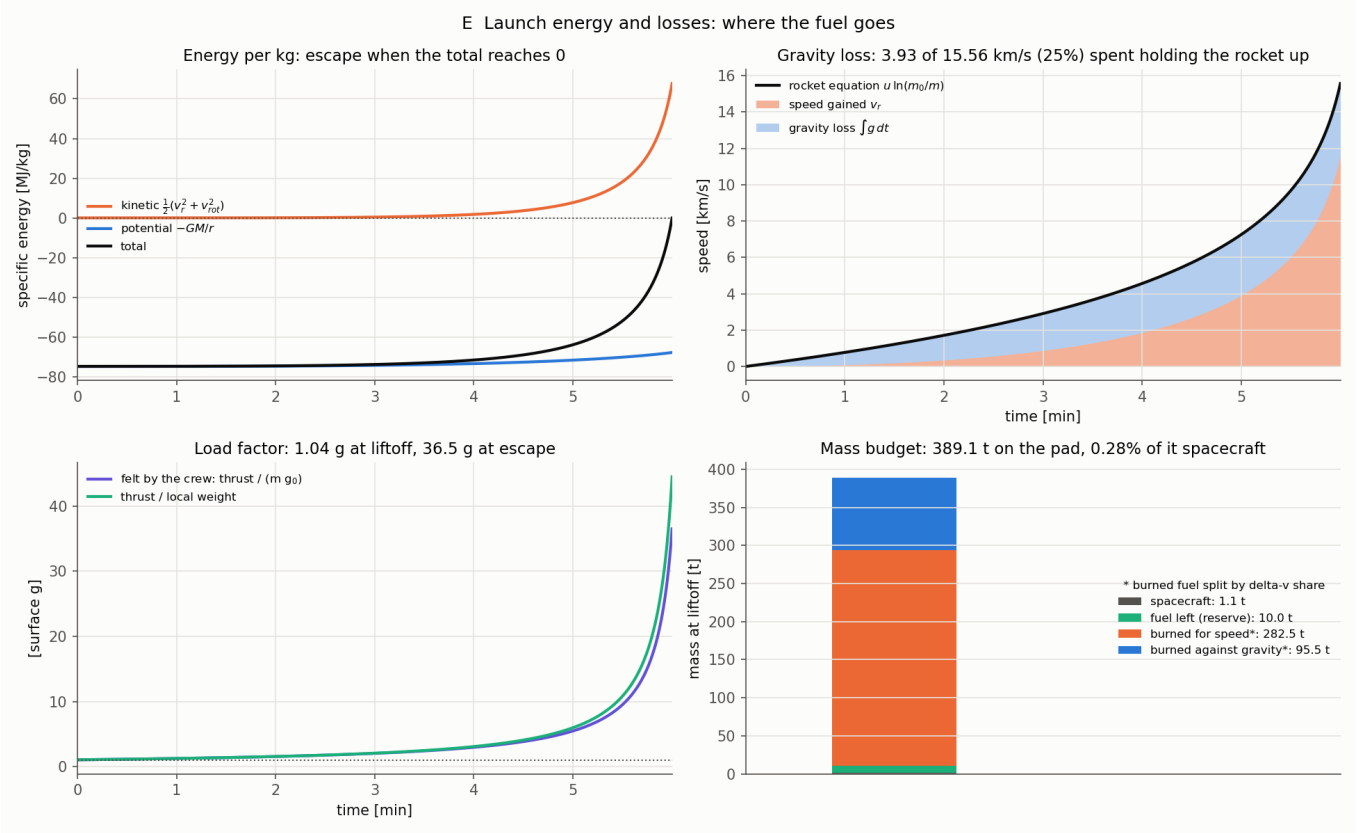


FIG. 16. Where the fuel goes. Energy per kg reaches zero at escape. The rocket equation's ideal Δv splits into speed gained and gravity loss. Load factor felt by the crew. Mass budget on the pad.

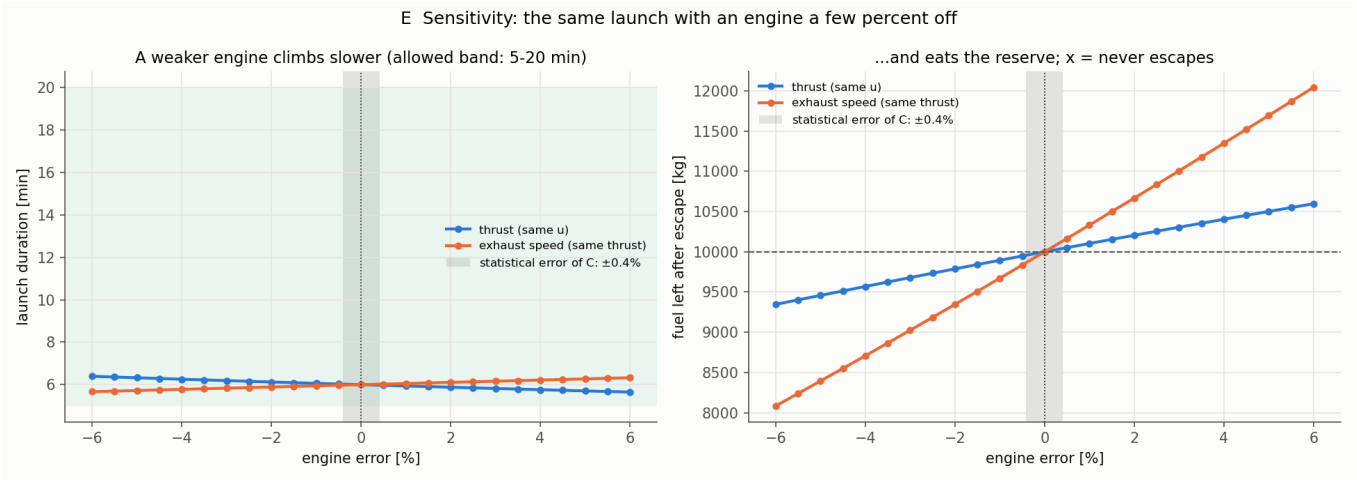


FIG. 17. The same launch with an engine that is weaker or stronger than measured. The grey band is the statistical uncertainty of our engine simulation.

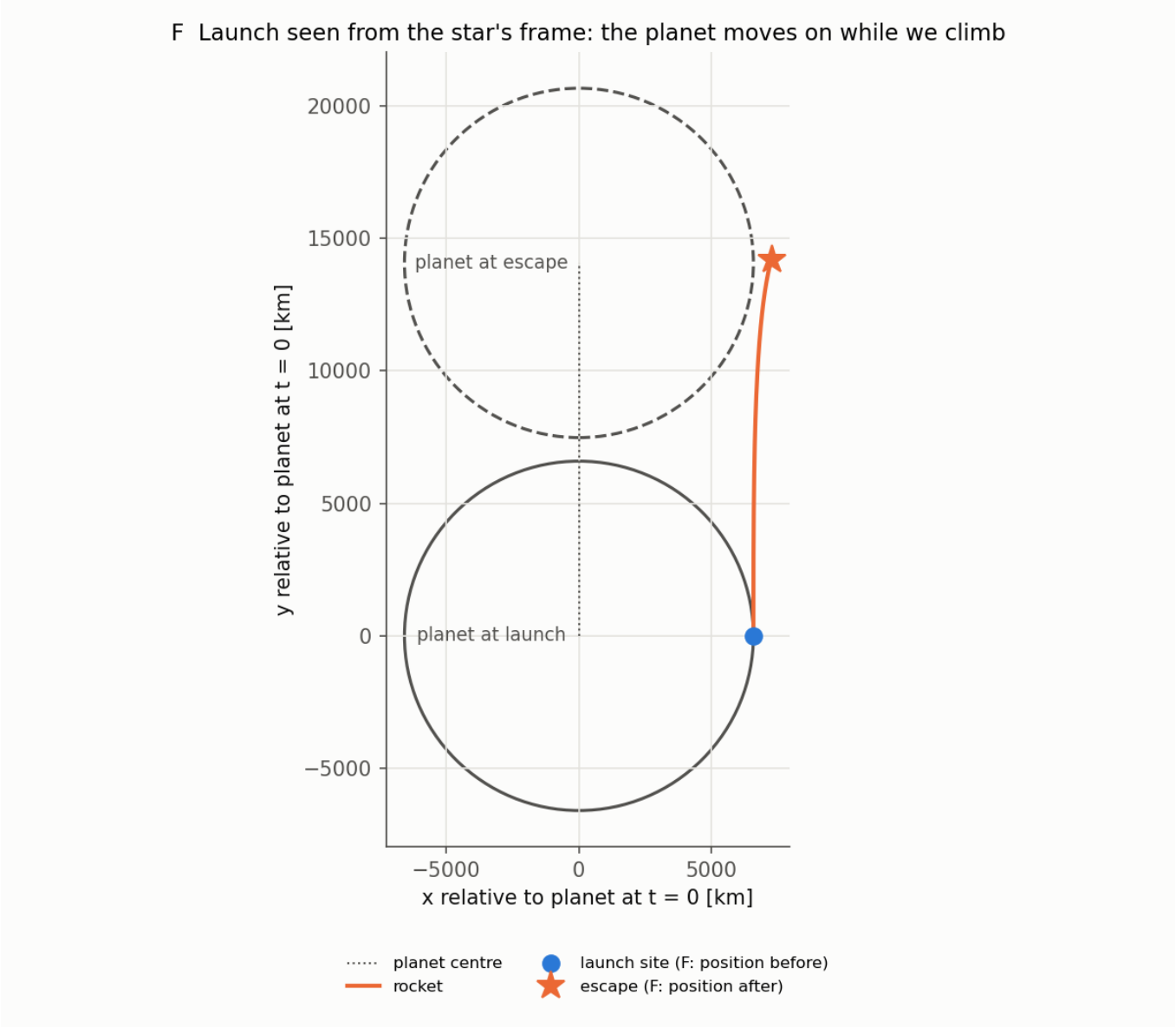


FIG. 18. The launch seen from the star's frame, relative to the planet's position at $t = 0$: the planet moves on while the rocket climbs.

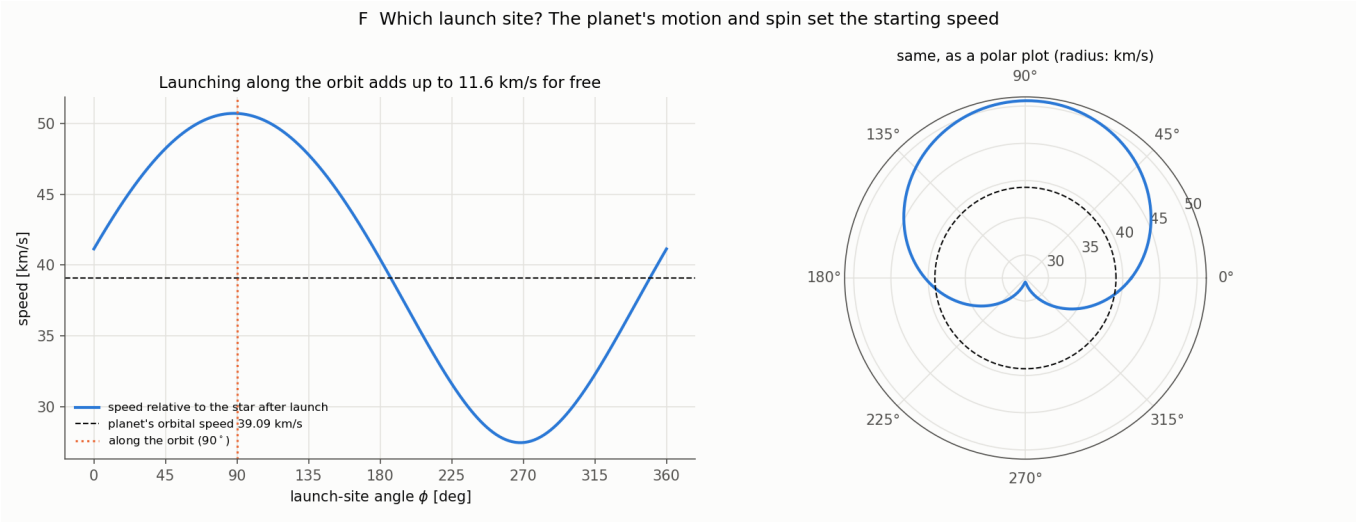


FIG. 19. Speed relative to the star right after launch, against the launch-site angle on the equator.